1/29/19.

Lecture 5.

Specific heats : Amount of heat reqd to raise a unit mass by 1° K. If a compressible substance, we can add heat in 2 ways : isobarically or isochorically.

By defn

$$C_p \equiv \left.\frac{\partial h}{\partial T}\right|_p$$

$$C_v = \left.\frac{\partial e}{\partial T}\right|_v$$

By defn

$$h \equiv e + \underbrace{pv}_{\text{flow work}}.$$

for ideal gas, TPG $\Rightarrow$ $h = f(T)$ only

$e = f(T)$ only.

$$C_p = \left.\frac{dh}{dT}\right|_p \; ; \; C_v = \left.\frac{de}{dT}\right|_v$$

For CPG, (calorically perfect gas)

C_p & C_p are constants.

$$h = C_p T$$

$$e = C_v T$$

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$$C_p T = C_v T + p v$$

$$C_p = C_v + \underbrace{\frac{p v}{T}}_{\text{Ideal gas law (IGL)} \Rightarrow pv = RT}$$

$$\Rightarrow \boxed{C_p = C_v + R} \qquad R = \frac{\mathbb{R}}{\mathcal{M}} = 287 \text{ J/K/kg for air @ 300K}$$

$$\gamma = \frac{C_p}{C_v} = \frac{n+2}{n} \; ; \; n = \#\text{ of DOF of molec.}$$

For atomic species : only translation in x, y, z dir $\Rightarrow$ 3 DOF

$$\gamma = \frac{3+2}{3} = \frac{5}{3}$$

$$= 1.67 \text{ for atomic.}$$

For diatomic : 3 translation, 2 rotation : 5 DOF

$$\gamma = \frac{7}{5} = 1.4 \text{ for diatomic}$$

For polyatomic, have more DOF in vib. ; $n = 7$

$$\gamma = \frac{9}{7} = 1.28.$$

$\rightarrow$ This is all @ STP.

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In genl., C_p & C_v are $f(T)$

For small ΔT, can assume CPG (what is small?)

At higher T, γ decreases

For air, $\gamma = 1.4$ up till $\approx$ 800-1000K & decreases to 1.2 @ T = 3500K

Governing Eq^{n}s

Can write any conservation eq^n in the form

Convection + accumulation + diffusion = production

↓ fluid motion

↓ time dependent

↓ by gradient

We will only consider steady state solutions $\Rightarrow$ accumulation is always zero.

Can write conservation eq^{n}s for

mass (continuity)

momentum (along 3 axes, Navier Stokes Eq^n, Newton's 2nd)

Energy (1st law of t/d)

Species (1 eq^n per species) (can be a large # of eq^n)

Will also need equation of state: most likely will be IGL.

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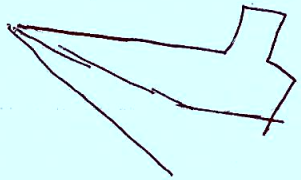
In genl., we have no analytical sol^{n}s. So we need to simplify the problem.



- Flow is primarily in one dimension. So assume 1D flow.
- Most of what we are interested in is moving axially. So we will need to account for such things as fuel injection, mixing, combustion & B.Ls, all of which are multidimensional effects.
- An appropriate way to average in the cross-stream direction is called "stream thrust averaging".
  - → Assumes that fluid props are perfectly uniform @ each axial location. Flow is uniformly mixed & has no gradients
  - → Assume flow is in local thermodynamic eq^m (LTE). This means that our state postulate applies @ any given location. We can define local t/d variables.
  - → Since we are assuming uniform properties @ any axial location, there cannot be any cross stream diffusion.
  - → Since the velocities are very high axially, axial convection will be important
    - → Axial diffusion is assumed to be very small compared to axial convection, even though there are gradients in axial dir.
    - → Diffusion can be neglected
      - ⇒ simplifies very much;

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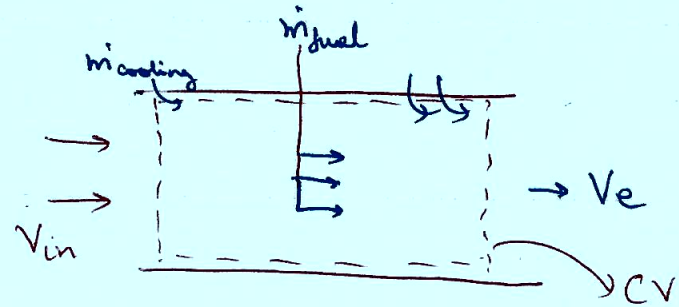
So governing eq^{n}s.

Convection = production

Continuity



Prod^n of mass = 0

$$\int_{CV} \rho \bar{V} \cdot d\bar{A} = 0$$

or w/ 1D approx.

$$\sum_i \rho V A = \sum_e \rho V A$$

For a vehicle, will need to account for fuel injection & transpiration cooling (convective cooling).
Typically we have single inlet & exit

Lump $\dot{m}_{fuel}$ & $\dot{m}_{cooling}$ in $\sum \dot{m}_b$ ↳ in their boundary.

So continuity becomes

$$\boxed{\rho_{in} V_{in} A_{in} + \sum \dot{m}_b = \rho_e V_e A_e}$$

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